Byung-Doh Oh

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Professional Website
GitHub Profile
Google Scholar

Research Interests

- Computational Linguistics; Psycholinguistics; Cognitive Modeling; Natural Language Processing; Large Language Models; Neural Network Interpretability
- My work aims to advance our understanding of real-time language processing in humans and machines by drawing on methods from computational psycholinguistics and machine learning.

Employment

Assistant Professor — Jan. 2026–
Linguistics and Multilingual Studies, Nanyang Technological University — Singapore

Assistant Professor/Faculty Fellow — Sept. 2024–Nov. 2025
Center for Data Science, New York University — New York, NY

Natural Language & Speech Processing Research Intern — May 2021–Aug. 2021
Tencent AI Lab — Bellevue, WA (remote)

- Mentor: Lifeng Jin

Education

The Ohio State University — Columbus, OH
Ph.D. in Linguistics — 2024

- Dissertation: *Empirical Shortcomings of Transformer-Based Large Language Models as Expectation-Based Models of Human Sentence Processing*
- Advisor: William Schuler
- Committee: Michael White, Micha Elsner, Tal Linzen

Seoul National University — Seoul, Korea
M.A. in English Language Education — 2018

- Thesis: *Exploring English Online Research and Comprehension Strategies of Korean College Students*
- Advisor: Youngsoon So

Seoul National University — Seoul, Korea
B.A. in English Language Education, *summa cum laude* — 2016

- Thesis: *A Study on the Assessment Techniques of Language MOOCs*
- Advisor: Youngsoon So

Awards and Honors

Selection as DARPA Riser — 2022
Defense Advanced Research Projects Agency

Fulbright Graduate Study Award — 2018–2020
Bureau of Educational and Cultural Affairs

Chunjae Education Group Scholarship — 2015
Chunjae Education Group

College of Education Alumni Association Scholarship — 2011
Seoul National University

Preprints and Work in Progress

2026 Sathvik Nair* and **Byung-Doh Oh***. Clozing the gap: Exploring why language model surprisal outperforms cloze surprisal. To appear in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*.

2025 **Byung-Doh Oh** and Tal Linzen. To model human linguistic prediction, make LLMs less superhuman. Accepted with minor revisions to *Trends in Cognitive Sciences*.

2025 William Timkey, Kuan-Jung Huang, **Byung-Doh Oh**, Grusha Prasad, Suhas Arehalli, Tal Linzen, and Brian Dillon. Eye movements reveal a dissociation between prediction and structural processing in language comprehension. *PsyArXiv preprint*, eq2ra_v1.

Peer-Reviewed Articles and Proceedings Papers

2025 Christian Clark, **Byung-Doh Oh**, and William Schuler. How well does first-token entropy approximate word entropy as a psycholinguistic predictor? In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 47–57.

2025 **Byung-Doh Oh** and William Schuler. The impact of token granularity on the predictive power of language model surprisal. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 4150–4162.

2025 **Byung-Doh Oh**, Hongao Zhu, and William Schuler. The inverse scaling effect of pre-trained language model surprisal is not due to data leakage. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 1820–1827.

2025 **Byung-Doh Oh** and William Schuler. Dissociable frequency effects attenuate as large language model surprisal predictors improve. *Journal of Memory and Language*, *143*:104645.

2025 Christian Clark, **Byung-Doh Oh**, and William Schuler. Linear recency bias during training improves Transformers' fit to reading times. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 7735–7747.

2024 **Byung-Doh Oh** and William Schuler. Leading whitespaces of language models' subword vocabulary pose a confound for calculating word probabilities. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 3464–3472.

2024 **Byung-Doh Oh**, Shisen Yue, and William Schuler. Frequency explains the inverse correlation of large language models' size, training data amount, and surprisal's fit to reading times. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2644–2663.

2023 **Byung-Doh Oh** and William Schuler. Transformer-based language model surprisal predicts human reading times best with about two billion training tokens. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1915–1921.

2023 **Byung-Doh Oh** and William Schuler. Token-wise decomposition of autoregressive language model hidden states for analyzing model predictions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 10105–10117.

2023 **Byung-Doh Oh** and William Schuler. Why does surprisal from larger Transformer-based language models provide a poorer fit to human reading times? *Transactions of the Association for Computational Linguistics*, *11*:336–350.

2022 **Byung-Doh Oh** and William Schuler. Entropy- and distance-based predictors from GPT-2 attention patterns predict reading times over and above GPT-2 surprisal. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 9324–9334.

2022 **Byung-Doh Oh**, Christian Clark, and William Schuler. Comparison of structural parsers and neural language models as surprisal estimators. *Frontiers in Artificial Intelligence*, *5*:777963.

2021 Lifeng Jin, **Byung-Doh Oh**, and William Schuler. Character-based PCFG induction for modeling the syntactic acquisition of morphologically rich languages. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 4367–4378.

2021 Evan Jaffe, **Byung-Doh Oh**, and William Schuler. Coreference-aware surprisal predicts brain response. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 3351–3356.

2021 **Byung-Doh Oh**, Christian Clark, and William Schuler. Surprisal estimators for human reading times need character models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*, pages 3746–3757.

2021 **Byung-Doh Oh** and William Schuler. Contributions of propositional content and syntactic category information in sentence processing. In *Proceedings of the 11th Workshop on Cognitive Modeling and Computational Linguistics*, pages 241–250.

2021 **Byung-Doh Oh**. Team Ohio State at CMCL 2021 shared task: Fine-tuned RoBERTa for eye-tracking data prediction. In *Proceedings of the 11th Workshop on Cognitive Modeling and Computational Linguistics*, pages 97–101.

2019 **Byung-Doh Oh***, Pranav Maneriker*, and Nanjiang Jiang*. THOMAS: The hegemonic OSU morphological analyzer using seq2seq. In *Proceedings of the 16th Workshop on Computational Research in Phonetics, Phonology, and Morphology*, pages 80–86.

2019 Micha Elsner, Andrea D. Sims, Alexander Erdmann, Antonio Hernandez, Evan Jaffe, Lifeng Jin, Martha Booker Johnson, Shuan Karim, David L. King, Luana Lamberti Nunes, **Byung-Doh Oh**, Nathan Rasmussen, Cory Shain, Stephanie Antetomaso, Kendra V. Dickinson, Noah Diewald, Michelle McKenzie, and Symon Stevens-Guille. Modeling morphological learning, typology, and change: What can the neural sequence-to-sequence framework contribute? *Journal of Language Modelling*, *7*(1):53–98.

2018 **Byung-Doh Oh** and Youngsoon So. Exploring English online research and comprehension strategies of Korean college students. *English Teaching*, *73*(3):53–76.

2017 **Byung-Doh Oh**. Predicting L2 writing proficiency with computational indices based on n-grams. *Foreign Language Education Research*, *21*:1–20.

Edited Volumes and Proceedings

2026 **Byung-Doh Oh**, Tatsuki Kuribayashi, Giulia Rambelli, Ece Takmaz, Philipp Wicke, Jixing Li, and Ryo Yoshida. *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics.*

2025 Tatsuki Kuribayashi, Giulia Rambelli, Ece Takmaz, Philipp Wicke, Jixing Li, and **Byung-Doh Oh**. *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics*.

Invited Talks

2026 **Byung-Doh Oh**. *To model human linguistic prediction, make LLMs less superhuman.* Invited talk at the Graduate Institute of Linguistics, National Taiwan University, Taipei, Taiwan.

2025 **Byung-Doh Oh**. *Language model surprisal does not underpredict garden path effects in early eye-tracking measures.* Invited talk at the Laboratory for Computation and Language in Minds and Brains, Stanford University, Stanford, CA (online).

2025 **Byung-Doh Oh**. *Unveiling the language processing of humans and machines.* Invited talk at the Division of Linguistics and Multilingual Studies, Nanyang Technological University, Singapore.

2024 **Byung-Doh Oh**. *What can linguistic data tell us about the predictions of large language models?* Invited talk at the Department of Computer Science and Engineering and Graduate School of Artificial Intelligence, POSTECH, Pohang, Korea.

2024 **Byung-Doh Oh**. *The bigger-is-worse effects of model size and training data of large language model surprisal on human reading times.* Invited talk in the Distinguished Speakers in Language Science Colloquium Series, Saarland University, Saarbrücken, Germany.

2023 **Byung-Doh Oh**. *The bigger-is-worse effects of model size and training data of large language model surprisal on human reading times.* Invited talk at the Computation and Psycholinguistics Lab, New York University, New York, NY.

2022 **Byung-Doh Oh**. *Computational models of sentence processing and syntactic acquisition.* Invited talk at the Department of English, Dongguk University, Seoul, Korea (online).

Conference Presentations

2026 **Byung-Doh Oh** and Tal Linzen. *To model human linguistic prediction, make LLMs less superhuman.* Lightning talk presented at the 35th Joint Workshop on Linguistics and Language Processing, Seoul, Korea.

2026 **Byung-Doh Oh** and Tal Linzen. *To model human linguistic prediction, make LLMs less superhuman.* Poster presented at the 39th Annual Conference on Human Sentence Processing, Cambridge, MA.

2026 Sathvik Nair and **Byung-Doh Oh**. *Should language models replace the cloze task forever?* Poster presented at the 39th Annual Conference on Human Sentence Processing, Cambridge, MA.

2026 William Timkey, Kuan-Jung Huang, **Byung-Doh Oh**, Grusha Prasad, Suhas Arehalli, Tal Linzen, and Brian Dillon. *Prediction and structural processing dissociate: A large-scale eyetracking study.* Poster presented at the 39th Annual Conference on Human Sentence Processing, Cambridge, MA.

2026 Christian Clark, **Byung-Doh Oh**, and William Schuler. *Word-level estimation of Shannon and Rényi entropy improves reading time predictions.* Poster presented at the 39th Annual Conference on Human Sentence Processing, Cambridge, MA.

2025 **Byung-Doh Oh** and William Schuler. *Word frequency modulates the effects of model size and training data amount on language model surprisal.* Poster presented at the 38th Annual Conference on Human Sentence Processing, College Park, MD.

2025 **Byung-Doh Oh** and William Schuler. *Correcting language model word probabilities reveals a greater divergence between surprisal and human reading times.* Poster presented at the 38th Annual Conference on Human Sentence Processing, College Park, MD.

2025 Christian Clark, **Byung-Doh Oh**, and William Schuler. *Effects of recency bias on Transformers' predictions of reading times.* Talk presented at the 38th Annual Conference on Human Sentence Processing, College Park, MD.

2023 **Byung-Doh Oh** and William Schuler. *On the bigger-is-worse nature of pre-trained language model surprisal.* Poster presented at the 36th Annual Conference on Human Sentence Processing, Pittsburgh, PA.

2023 **Byung-Doh Oh** and William Schuler. *Memory-based predictors from GPT-2 attention predict reading times over surprisal.* Poster presented at the 36th Annual Conference on Human Sentence Processing, Pittsburgh, PA.

2022 **Byung-Doh Oh**. *Unified unsupervised grammar induction for typologically diverse languages.* Poster presented at the DARPA Risers program, Columbus, OH.

2021 **Byung-Doh Oh**, Christian Clark, and William Schuler. *Comparison of structural and neural language models as surprisal estimators.* Short talk presented at the 34th Annual CUNY Conference on Human Sentence Processing, Philadelphia, PA (online).

2021 **Byung-Doh Oh** and William Schuler. *Contributions of propositional content and syntactic categories in sentence processing.* Short talk presented at the 34th Annual CUNY Conference on Human Sentence Processing, Philadelphia, PA (online).

2019 Evan Jaffe and **Byung-Doh Oh**. *The role of learnability in morphological change: A computational approach.* Talk presented at the Fourth American International Morphology Meeting, Stony Brook, NY.

Teaching

SP26 **Instructor of Record**, HG4015/7015: Psycholinguistics, Nanyang Technological University

AU25 **Instructor of Record**, DS-GA 1006: Capstone Project and Presentation, New York University
(Co-taught with Aahlad Puli and Shauli Ravfogel)

2025 **Guest Lecture**, DS-GA 1012: Natural Language Understanding and Computational Semantics, New York University (Instructor: Tal Linzen)
"Unveiling the language processing of humans (and machines)"

2025 **Guest Lecture**, HG2051: Language and the Computer, Nanyang Technological University (Instructor: Hiram Ring)
"Treating text documents like pizza"

SP25 **Instructor of Record**, DS-GA 1015: Text as Data, New York University

2024 **Guest Lecture**, DS-GA 3001: Computational Linguistics and Cognitive Science, New York University (Instructor: Tal Linzen)
"Transformer-based language model surprisal predicts human reading times best with about two billion training tokens"

2024 **Guest Lecture**, Metro Early College High School
"Guessing meaning and breaking it down (with NACLO problems)"

AU20, SP21 **Instructor of Record**, LING 3801: Codes and Code Breaking, The Ohio State University

SU17 **Teaching Assistant**, College English 1, Seoul National University

Advising

2026– **Eric Tu**, M.A. student, Nanyang Technological University (Co-advised with Zoe Xi Chen)

Service

Organizing

2024– Workshop on Cognitive Modeling and Computational Linguistics

Reviewing

2026 ACL Rolling Review (Area Chair), CogSci 2026, CoNLL 2026, *Journal of Memory and Language*, *Nature Communications*, *Transactions of the Association for Computational Linguistics*

2025	ACL Rolling Review, CogSci 2025, PACLIC 2025, *Cognition*, *Communications Psychology*, *Computational Linguistics*, *Korean Journal of Linguistics*, *Nature Human Behavior*
2024	ACL Rolling Review, BlackboxNLP, CogSci 2024, *eLife*, *Frontiers in Communication*, *Journal of Memory and Language*, *Nature Computational Science*
2023	ACL Rolling Review
2022	ACL Rolling Review
2021	ACL Rolling Review, CMCL 2021

Departmental/University

2026	Search Committee, Assistant Professor in Digital Humanities, NTU
2026–	Postgraduate Committee, Linguistics and Multilingual Studies, NTU
2026–	Development Committee, Linguistics and Multilingual Studies, NTU
2024–2025	MS Admissions Committee, Center for Data Science, NYU
2019–2024	Laboratory/Computing Committee, Department of Linguistics, OSU
2019–2020	Treasurer, Student Linguistics Association, OSU
2018–2019	Speakers Committee, Department of Linguistics, OSU
2017–2018	Resident Advisor, Gwanak Residence Halls, SNU
2015–2017	International Student Assistant, Gwanak Residence Halls, SNU

Other

2020, 2022	Student Volunteer, North American Computational Linguistics Olympiad (NACLO)
2015–2016	K-MOOC Monitoring Team, National Institute for Lifelong Education, Seoul, Korea
2012–2014	Translator/Interpreter (ROK Air Force), ROK/US Combined Forces Command, Seoul, Korea

Skills

Languages	Korean (native), English (fluent), Japanese (conversational)
Programming Languages	Python, C++, Bash, R, LaTeX, Prolog
Frameworks	PyTorch, TensorFlow, HuggingFace, pandas, Matplotlib, Git, Slurm, PBS